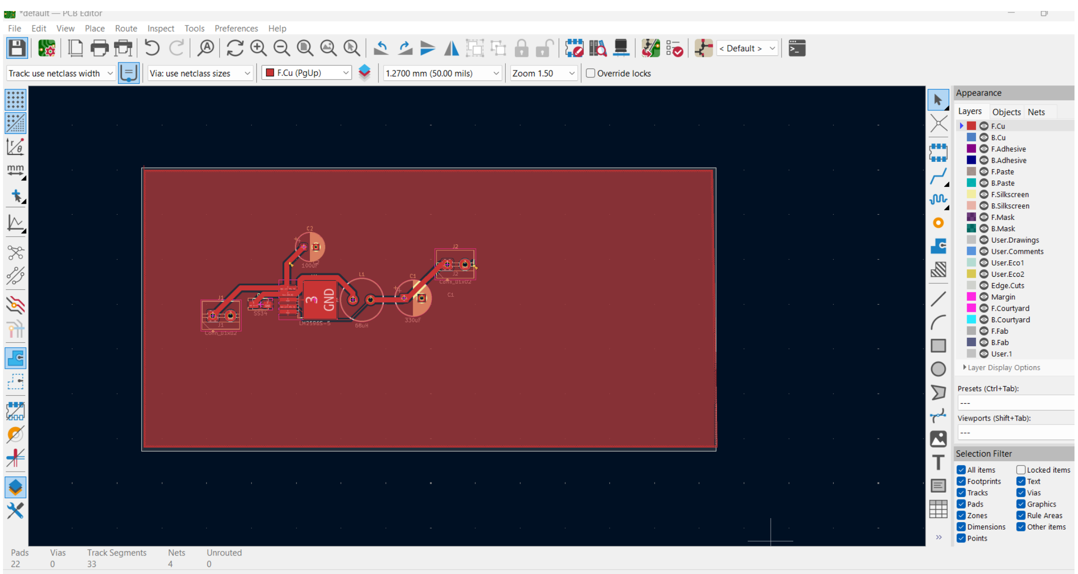
Fortune Ngwenya
Buck Convertor Project

12V to 5V DC-DC Converter Design, LTspice Simulation, KiCad PCB Layout, and Manufacturing Readiness

---

Portfolio Engineering Report

A compact power-electronics project demonstrating converter theory, transient simulation, waveform verification, schematic capture, PCB routing, ground-plane implementation, and design-rule checking.



Final routed PCB view with copper pour / ground-plane visualization.

1. Executive Summary

This project presents a 12V-to-5V buck converter workflow from circuit concept through simulation and PCB implementation. The converter was first modeled in LTspice using a switch-level buck topology to verify fundamental switching behavior, output regulation trend, inductor current ripple, startup response, and switch-node waveform. The design was then implemented in KiCad using an LM2596S-5 regulator, SS34 Schottky diode, 68 µH inductor, input/output capacitors, connectors, and a copper ground plane.

Recruiter takeaway: This project demonstrates practical EE skills: power electronics, simulation, PCB design, layout review, design-rule checking, and manufacturing-file awareness.

Item	Target / Result
Input Voltage	12 V DC
Nominal Output Target	5 V DC
Load Target	≈ 2 A / 10 W class
Simulation Tool	LTspice
PCB Tool	KiCad
Regulator Used in PCB	LM2596S-5
Key Evidence	Waveforms, schematic, PCB layout, ground pour, DRC result

2. Design Objectives

The project objective was not only to draw a converter, but to document a complete engineering workflow: define specifications, simulate behavior, build the PCB schematic/layout, verify routing, and generate a portfolio-ready record of design decisions.

Objective	Evidence in Project
Step down 12V to approximately 5V	LTspice output startup and steady-state waveform
Show switching behavior	Switch-node waveform alternating between high and low states
Verify inductor energy transfer	Triangular inductor current ripple
Create a manufacturable PCB	KiCad schematic and PCB layout screenshots
Check physical layout integrity	DRC screenshot showing 0 errors

3. Buck Converter Theory and Calculations

An ideal buck converter steps down DC voltage using a controlled switch, diode/synchronous path, inductor, and output capacitor. For an ideal converter operating in continuous conduction mode, the duty cycle is approximately the output voltage divided by the input voltage.

Ideal duty cycle: D = Vout / Vin = 5 / 12 = 0.417 ≈ 41.7%

Output power target: Pout = Vout × Iout = 5 V × 2 A = 10 W

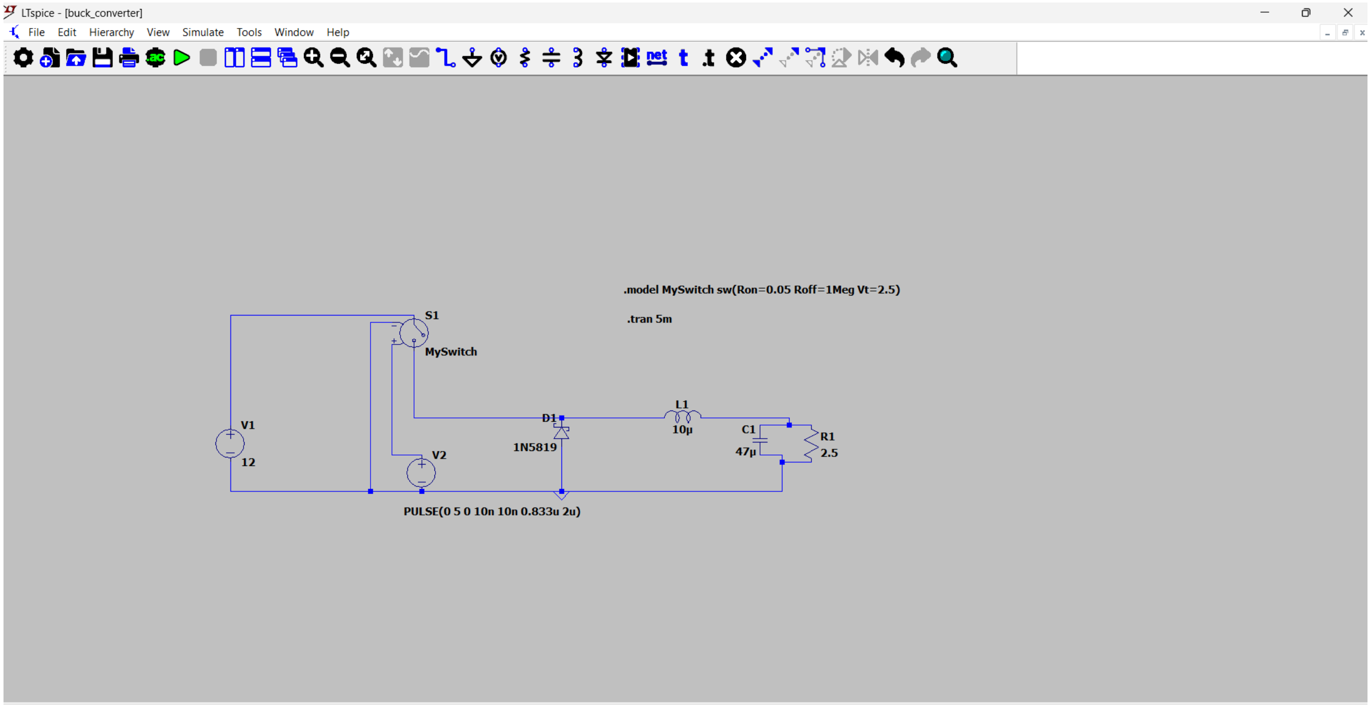
Equivalent load: Rload = Vout / Iout = 5 V / 2 A = 2.5 Ω

The LTspice switch command used a pulse drive approximating the calculated duty ratio. The switch-level simulation is useful for verifying buck-converter fundamentals before moving into a regulator-based PCB implementation.

Component	Value / Part	Purpose
V1	12 V	Input supply
Switch / LM2596 stage	Behavioral switch in LTspice; LM2596S-5 in PCB	Step-down conversion control
D1	1N5819 / SS34	Freewheel Schottky path
L1	10 µH in LTspice, 68 µH in KiCad PCB	Energy storage and current smoothing
Cout	47 µF in LTspice, 330 µF in PCB	Output ripple filtering
Load	2.5 Ω	Represents ~2 A output at 5 V

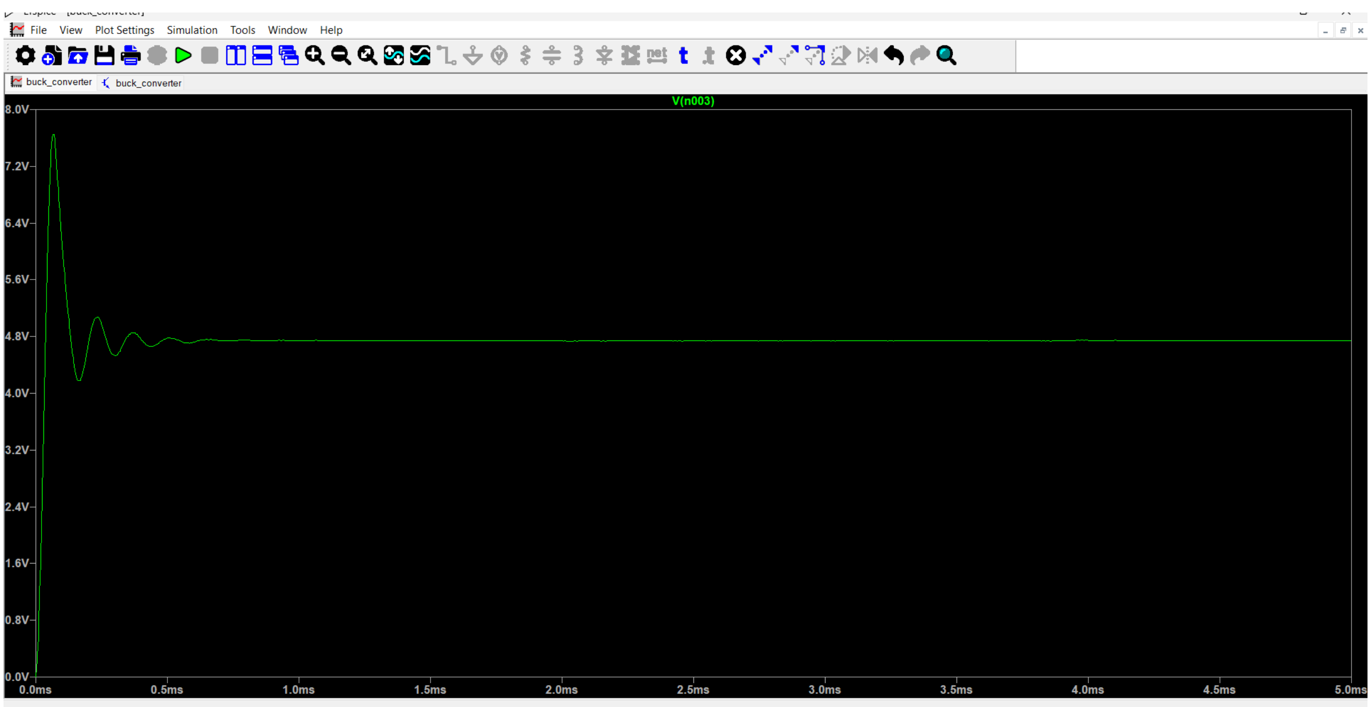
4. LTspice Simulation Evidence

4.1 LTspice Buck Converter Schematic



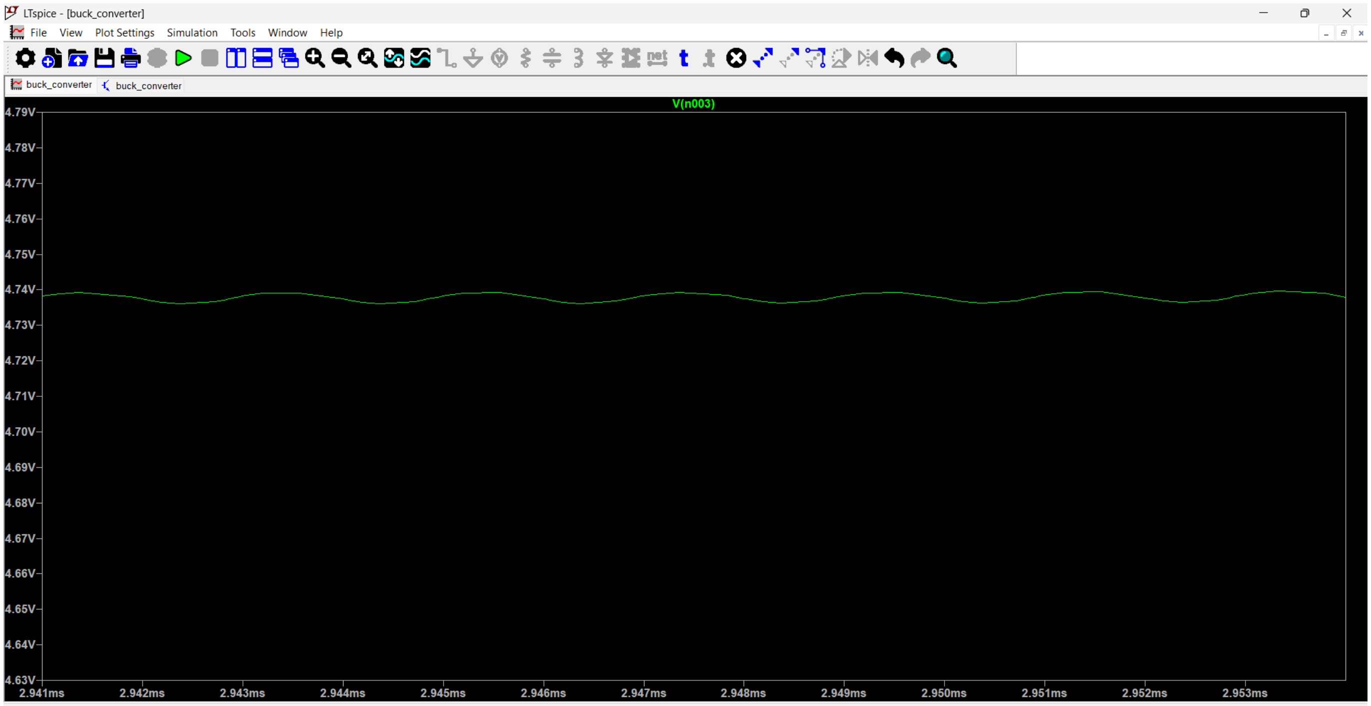
Switch-level LTspice model used to verify buck-converter operation before PCB layout. The pulse source implements the approximate duty cycle needed to step 12 V down toward 5 V.

4.2 Startup Response



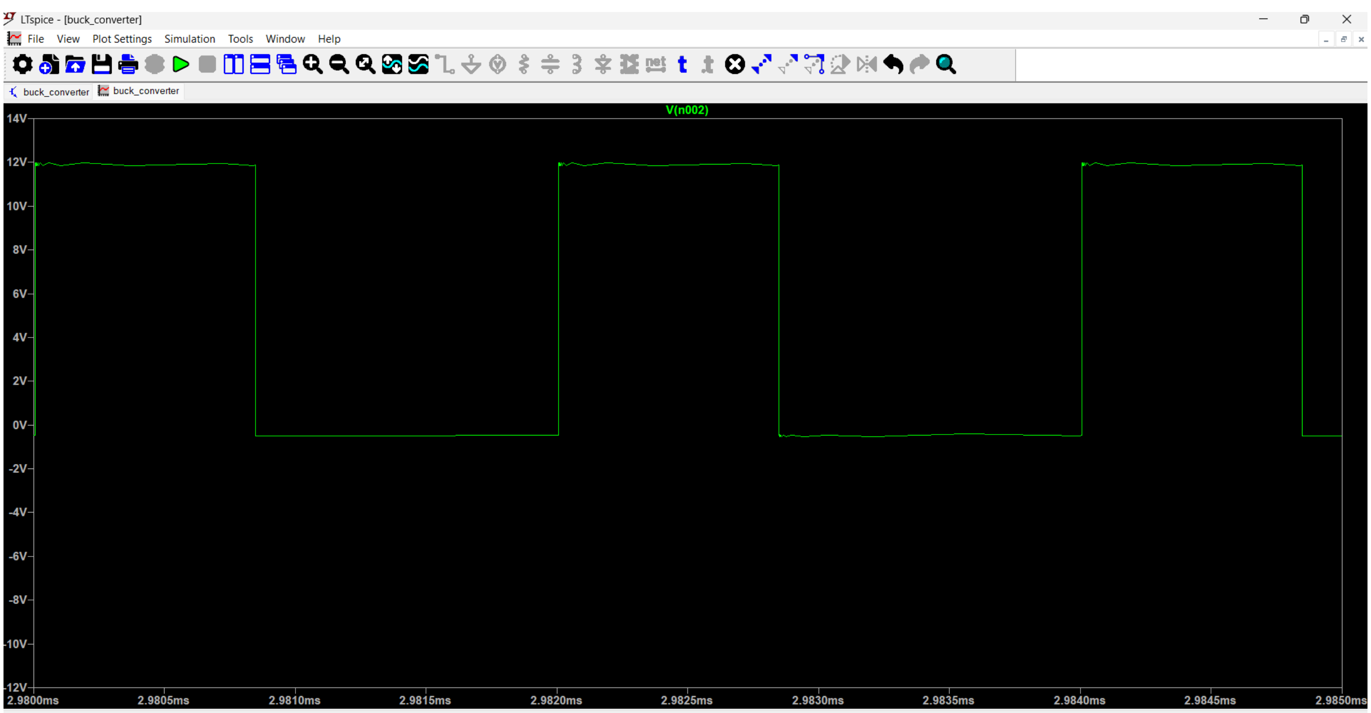
Output voltage startup waveform. The converter rises from 0 V, shows transient overshoot/ringing, and then settles near the regulated operating region.

4.3 Output Ripple at Steady State



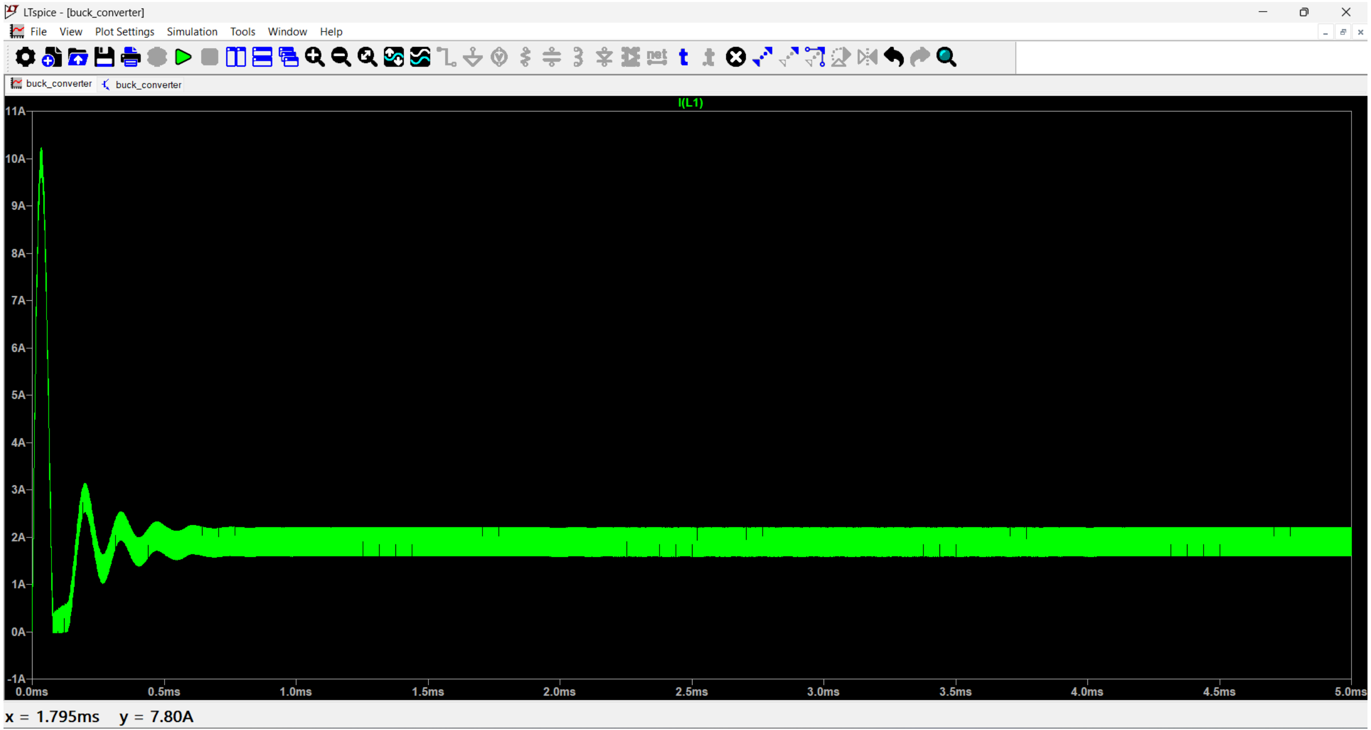
Zoomed output-voltage ripple near steady-state operation. The waveform shows small periodic ripple around approximately 4.74 V in the simplified LTspice model.

4.4 Switch Node Voltage



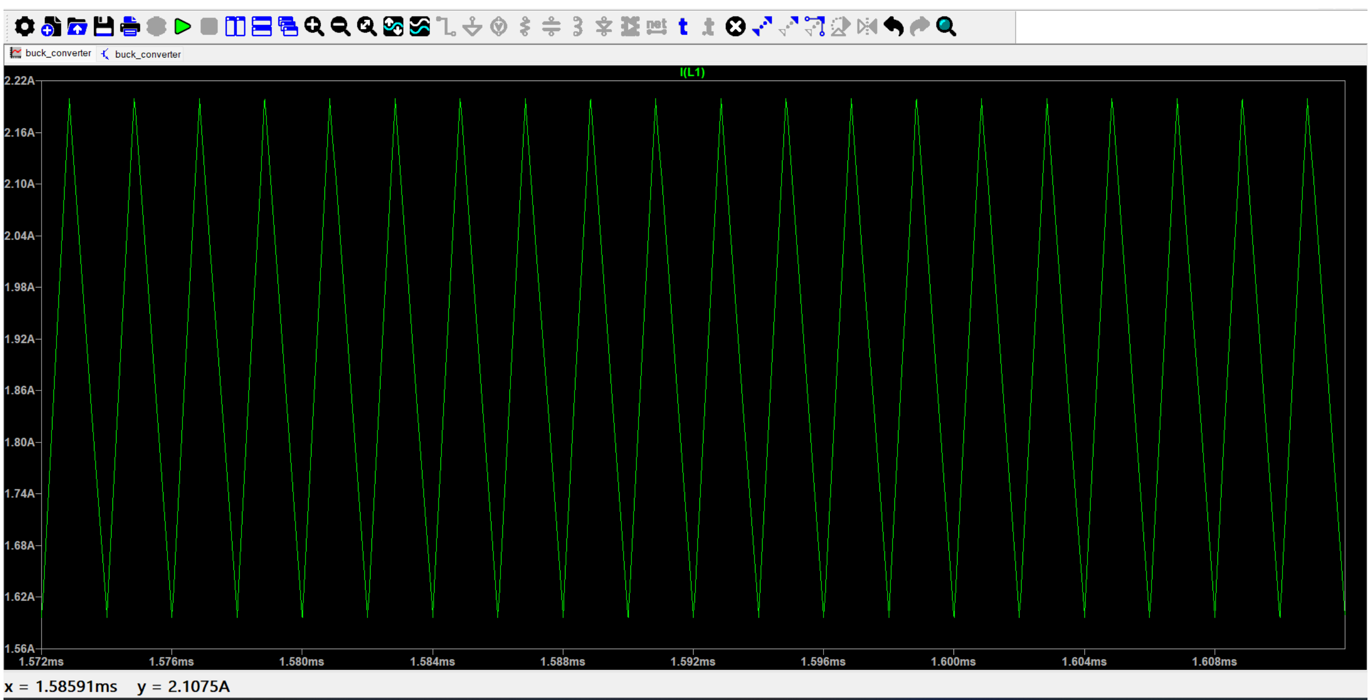
Switch-node waveform alternating between high and low states. This verifies the switching action that transfers energy through the inductor.

4.5 Inductor Current Startup



Inductor current startup waveform. The initial current surge and settling behavior show energy buildup in the inductor before reaching steady-state switching behavior.

4.6 Inductor Current Ripple



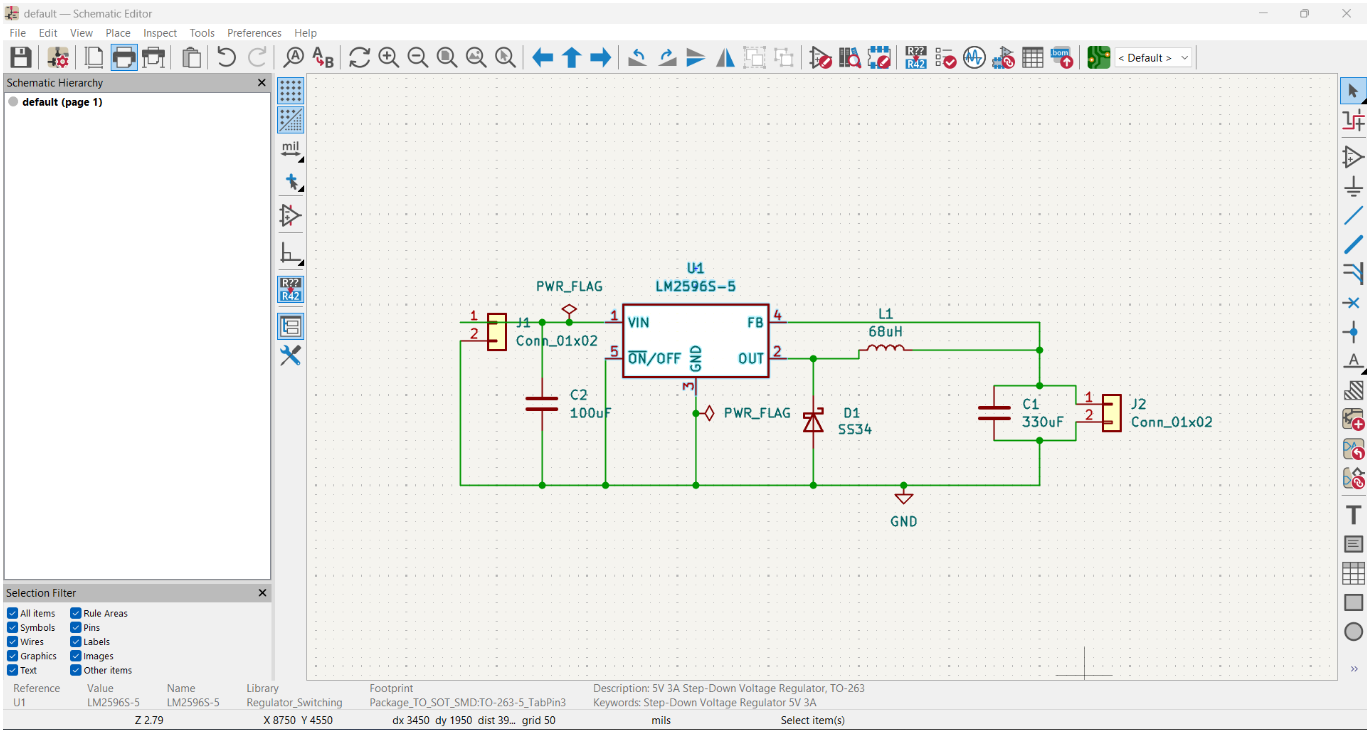
Steady-state inductor current ripple. The triangular waveform is characteristic of a buck converter operating in continuous conduction mode.

Simulation Interpretation

The waveform set verifies the core physics of a buck converter: the switch node toggles between supply and low voltage, the inductor current ramps up and down rather than dropping to zero, and the output capacitor smooths the pulsed energy into a nearly DC output. The simplified LTspice model settles near 4.74 V instead of an exact 5.00 V, which is reasonable for a switch-level model with idealized/non-IC feedback behavior and component simplifications.

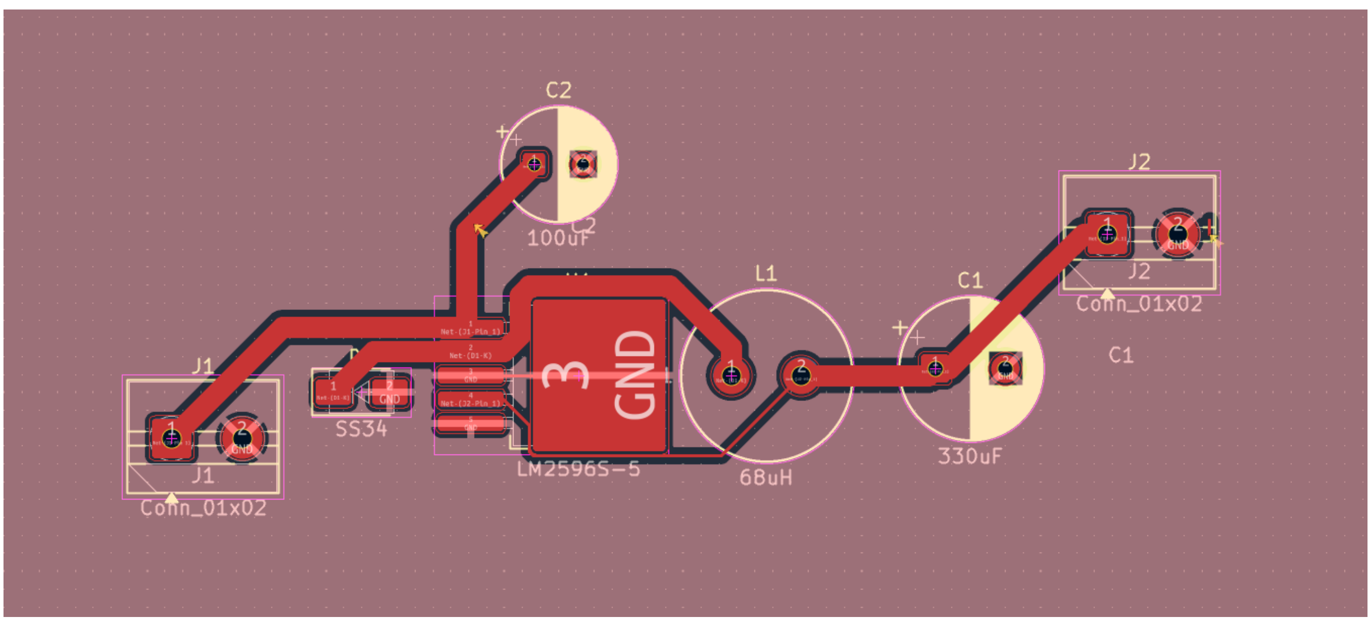
5. KiCad Schematic and PCB Implementation

5.1 KiCad Schematic Capture



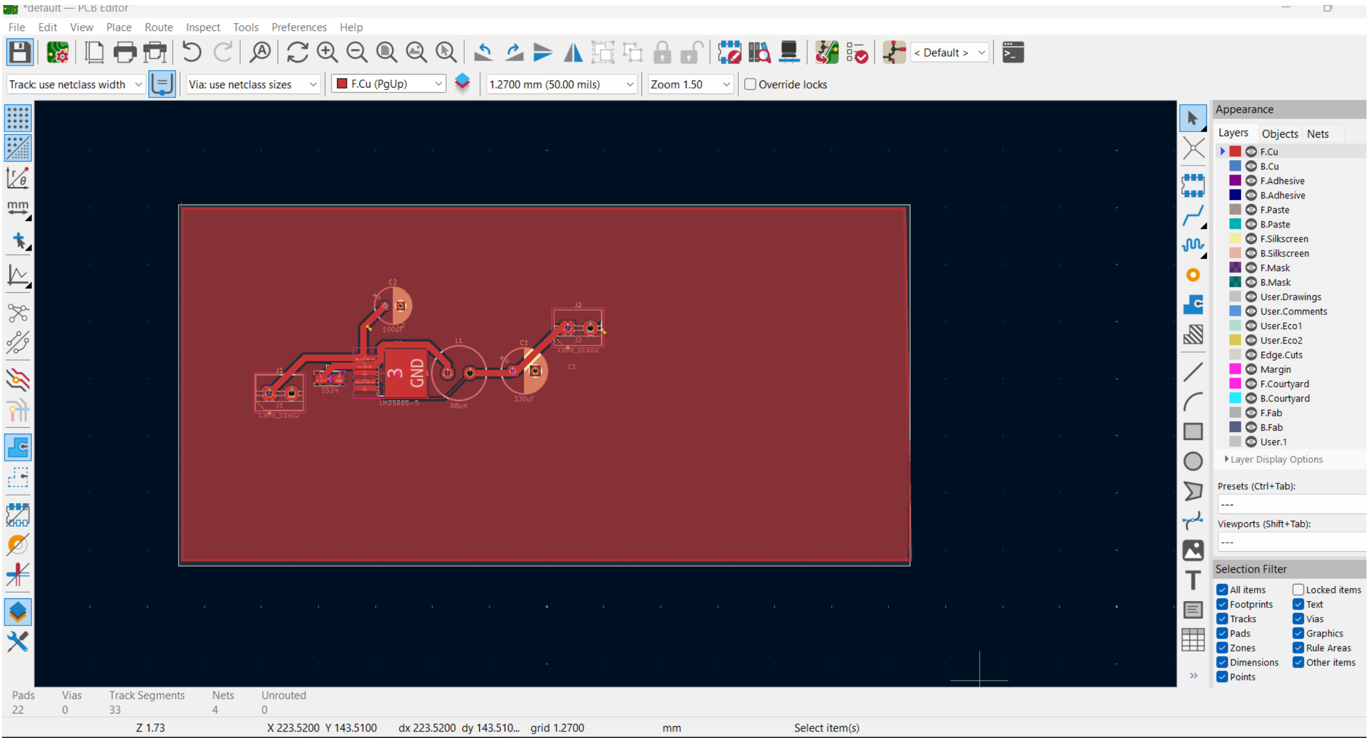
KiCad schematic implementation using the LM2596S-5 regulator, SS34 diode, 68 µH inductor, 100 µF input capacitor, 330 µF output capacitor, input connector, and output connector.

5.2 PCB Routing Layout



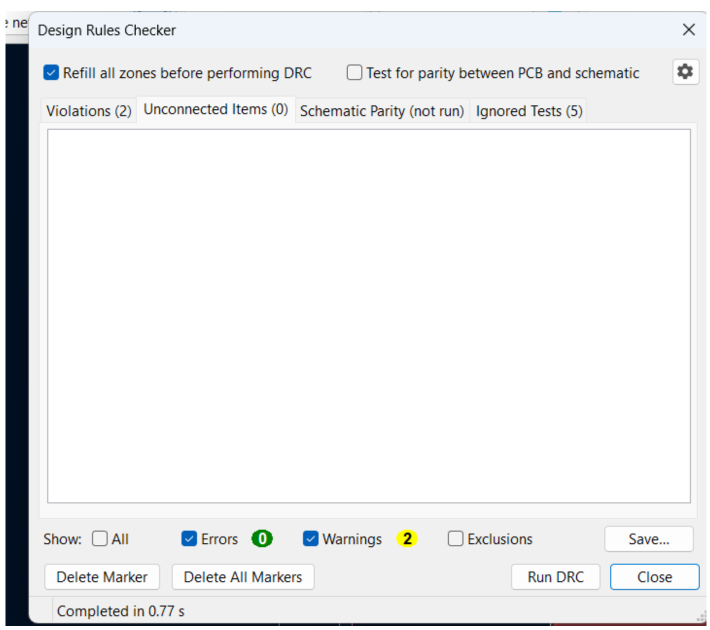
PCB layout showing routed power paths and component placement. The design places the regulator, diode, inductor, capacitors, and connectors into a compact switching-supply layout.

5.3 Ground Plane Implementation



PCB with copper ground pour / ground-plane visualization. A ground plane helps reduce return-path impedance and improves layout robustness for switching power designs.

5.4 Design Rule Check



KiCad Design Rules Checker result showing zero errors and zero unconnected items. Two warnings remain, but no critical routing errors were reported.

6. PCB Design Notes

For a switching regulator PCB, layout matters because fast current transitions can create noise, voltage spikes, and EMI. The most important routing considerations are short high-current paths, controlled switch-node area, low-impedance ground return, and close placement of input/output capacitors.

Design Choice	Why It Matters
Wide power routing	Reduces resistive loss and heating in high-current paths
Ground pour	Improves return path and lowers impedance
Input/output capacitors near regulator path	Helps reduce ripple and switching noise
DRC verification	Confirms manufacturability constraints and routing integrity
Gerber/drill outputs	Supports fabrication-ready workflow

7. Results Summary

Test / Evidence	Result
LTspice schematic	Switch-level converter model completed
Startup response	Output rises and settles after initial transient
Output ripple	Small steady-state ripple observed near output operating point
Inductor ripple	Triangular continuous-conduction current observed
Switch-node voltage	PWM-like high/low switching waveform verified
KiCad PCB layout	Board routed with ground pour
DRC	0 errors, 0 unconnected items

Project Conclusion

This project demonstrates a complete buck-converter workflow: simulation, waveform analysis, schematic capture, PCB layout, ground-plane implementation, and design-rule verification.

The design evidence includes:
• LTspice switching waveforms
• Startup and ripple analysis
• KiCad schematic implementation
• Routed PCB layout
• Ground-pour integration
• DRC validation

The project showcases practical power-electronics design, simulation, PCB-development, and technical documentation skills.